MIL-C-46621C(AR)
1 May 1978
SUPERSEDING
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30 September 1970

MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH EXPLOSIVE INCENDIARY, M97A2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electric primed cartridge for use in Gun, 20mm, Automatic, M24 and M24A1.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont)

MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

D7258928 - Marking on M548 Shipping and Storage Container for 20mm Cartridges
F7258967 - Method of Packing 150 20mm Cartridges, in M10 Links, Three 50-round Belts, in M548 Container, Packed for Single-loop Right-hand Feed and Double-loop Left-hand Feed
D7259063 - Cartridge, 20mm, High Explosive Incendiary, M97A2
LI 7259063 - Index of Inspection Equipment Lists for: Cartridge, 20mm, High Explosive Incendiary, M97A2
F7259119 - Method of Packing 200 Cartridges, 20mm, HEI and APT with M21A1 Type Cases, in Bulk, in M548 Container
F7259122 - Method of Bulk Packing 150 Cartridges, 20mm, HEI and APT, with M21A1 Type Cases, in M548 Container
F7259188 - Method of Packing 150 20mm Cartridges in M10 Links, Three 50-round Belts, in M548 Container, Packed for Single-loop Left-hand and Double-loop Right-hand Feed
F7259189 - Method of Packing 150 20mm Cartridges in M10 Links One Belt, in M548 Container, Packed for Single-loop, Left-hand and Double-loop Right-hand Feed
D7259237 - Marking on M548 Shipping and Storage Container for Cartridges, 20mm, Linked (Functional)
F11075755 - Method of Packing 150 20mm Cartridges, in M10 Links, One Belt, in M548 Container, Packed for Single-loop, Right-hand Feed and Double-loop Left-hand Feed

PUBLICATIONS

U.S. ARMY MATERIEL DEVELOPMENT AND READINESS COMMAND

DARCOM-R 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges (see 6.9)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. The cartridge shall be in accordance with Drawing (Dwg.) D7259063, referenced specifications, and other requirements specified herein.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.2)

3.3 First article sample. Unless otherwise directed by the contracting officer, a first article sample shall be required.

3.4 Projectile torque. The projectile shall withstand a minimum torque without rotary movement as specified by Dwg. D7259063.

3.5 Projectile extraction. The force necessary to extract the projectile from the cartridge case shall be as specified on Dwg. D7259063.

3.6 Electrical resistance, primed cases. The electrical resistance of the primed cases shall be not less than 500 ohms or more than 1,200,000 ohms.

3.7 Velocity. The average velocity of the sample cartridges conditioned at 68 degrees (°) Fahrenheit (F) to 72°F shall be 2680 ± 50 feet per second (fps) at 78 feet from the test weapon muzzle and the sample standard deviation shall not exceed 40 fps.

3.8 Pressure. The average chamber pressure ($\bar{X}$) of the sample cartridges, conditioned at 68°F to 72°F, plus 0.84 times the average range ($\bar{R}$) of the chamber pressure shall not exceed 51,000 pounds per square inch (psi).

3.9 Waterproofness. The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F to 72°F shall not vary from the average velocity (dry) by more than 100 fps.

3.10 Accuracy. The average mean radius of all targets of the sample cartridge projectiles at ambient temperature shall not exceed 15 inches at a range of 600 yards.

3.11 Action time. The action time of the sample cartridge, at ambient temperature, shall not exceed 4.0 milliseconds.

3.12 Projectile function. The projectile of the sample cartridge at ambient temperature shall function with high order detonation upon target impact.

3.13 Function and casualty. The cartridge, at ambient temperature, shall function without casualty.

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3.14 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.14.1 Process defects. The cartridge shall be free of defects such as folds, wrinkles, buckles, cracks, splits, severe scratches, cuts, dents, perforations, and other forms of substandard workmanship.

3.14.2 Cleanliness. The cartridge and its components shall be free of all foreign matter. (see 6.6)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 First article sample. The sample shall consist of 430 cartridges and shall be delivered for inspection in accordance with the first article approval provisions of the contract and MIL-A-2550. Identification shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection. The first article sample will be subjected to the quality conformance inspection provisions as stated herein, except that approval will be based upon the examination of the complete sample for visual and dimensional characteristics and the initial test sample in accordance with Table I. (see 6.3)

4.4 Quality conformance inspection.

4.4.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

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4.4.1.1 Lot. A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectile metal parts, fuze assemblies) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.4.1.2 Lot identification. Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a National Stock Number assigned by the procuring activity.

4.4.2 Examination. One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

4.4.2.1 Primed cartridge cases. Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.4.1. Primed cases which fail to meet the requirement of 3.6 shall be removed from the lot.

4.4.2.2 Sampling and acceptance criteria. The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.3.c) In addition, where three or less defects appear in a classification, or, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.4)

4.4.2.3 AQL's. The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

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4.4.2.3 (Cont)

c. Individual defects associated with the class basis:

Major ---------- 0.40%

4.4.3 Classification of defects 1/. The classification of defects shall be as follows:

4.4.3.1 Cartridge assembly, prior to application of wax coating (see Dwg. D7259063 and D7259064, a detail of Dwg. D7259063).

Category and Defect	Method of Inspection	
Critical:		
1. Split or perforated case	Visual	
2. Crack or split in projectile	Visual	
Major:		
101. Loose primer	Manual	
102. Depth of primer seating	Gage	
103. Overall length, max.	Gage	
104. Improper, incomplete or missing crimp (case-projectile)	Visual	
105. Profile and alignment, max.	Gage	2/
106. Corrosion	Visual	
107. Head configuration	Visual	
108. Metal sliver(s) on head face	Visual	
109. Crack, split or dent in fuze nose cap	Visual	
110. Fuze loose	Gage	3/
Minor:		
201. Marking incorrect, incomplete, illegible or missing	Visual	
202. Projectile protective coating damaged or incomplete	Visual	
203. Waterproofing missing around primer	Visual	
204. Scratch, dent, buckle, bulge, wrinkle or fold in case	Visual	
205. Rotating band damaged	Visual	
206. Primer crimp missing	Visual	
207. Foreign matter, except corrosion	Visual	
208. Incorrect type cartridge	Visual	
209. Gap between fuze and projectile body	Gage	
210. Lacquer on primer button	Visual	

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4.4.3.2 Cartridge assembly after application of wax coating (see Dwg. D7259063).

Category and Defect	Method of Inspection
Critical: None defined.	
Major:	
101. Wax coating missing or incomplete	Visual
Minor:	
201. Wax coating on projectile	Visual
202. Wax coating flaky or otherwise unsatisfactory	Visual

1/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

3/ When the fuze turns with respect to the projectile body upon application of the specified torque in a counterclockwise direction, the cartridge shall be classed defective.

4.4.4 Tests. The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges which have met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests. The combining of tests is permitted. (see 6.5)

Table I

Test 1/		Sample Size Production Lot	Rqmt Para
Projectile torque	2/	20	3.4
Projectile extraction	3/	20	3.5
Propellant contamination	4/	20	3.14.2
Electrical resistance verification	5/	20	3.6
Velocity	6/	20	3.7
Pressure	7/	20	3.8
Waterproofness	8/	20	3.9
Accuracy	9/	40	3.10
Action time	10/	50	3.11
Projectile function	11/	50	3.12
Function and casualty	12/	200	3.13

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4.4 4 (Cont)

1/ The lot shall be rejected if in any firing test one or more of the following malfunctions or casualties of Table II occur:

a. Misfire associated with e or f of Note 1/ of Table II.
b. Projectile remaining in bore.
c. Metal parts separation, except rotating band separation (see Table II, Note 5/).
d. Premature projectile function.

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

2/ Projectile torque. Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

3/ Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to comply with the maximum requirement.

4/ Propellant contamination. The presence of foreign matter in the propellant of one or more of the test cartridges will cause rejection of the lot. (see 3.14.2, 4.5.3 and 6.6)

5/ Electrical resistance verification. The lot will be rejected if any primer of the sample cartridgesfails to comply with either of the primer resistance requirements.

6/ Velocity. If the sample fails to comply with either or both requirements, a retest sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to comply with the applicable requirements. (see 6.7.1)

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4.4.4 (Cont)

7/ Pressure. If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 51,000 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirement. (see 6.7.2)

8/ Waterproofness. If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.7.

9/ Accuracy. If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 80 cartridges shall be tested. The lot shall be rejected if the average mean radius of eight retest accuracy targets exceeds the applicable requirements.

10/ Action time. If any sample cartridge fails to meet the requirement for action time, the lot shall be rejected.

11/ Projectile function. Failure of five or more sample cartridges to comply with the applicable requirement shall cause rejection of the lot. If more than two or less than five fail to comply, a second sample of 50 cartridges shall be tested. The lot shall be rejected if seven or more cartridges of the combined samples fail to comply with the applicable requirements.

12/ Function and casualty. The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

4.4.4.1 Firing defects. Firing defects and associated criteria shall be as specified in Table II. For defect definitions, see DARCOM-R 715-505, Volume 8. (see 6.9)

Table II

Firing Defects	Acc	Rej
Cartridge:		
Misfire 1/	- -	- -
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1

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4.4.4.1 (Cont)

Table II (Cont)

Firing Defects	Acc	Rej
Primer:		
Missing primer or button missing from primer	0	2
Primer leak	9	25
Case:		
Longitudinal split 2/:		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture (partial) 2/:		
S, J, or K	4	11
L	0	1
Circumferential rupture (complete) 2/	0	1
Detached metal 3/	0	1
Projectile:		
Rotating band separation:		
Complete or partial separation having band seat serration imprint 4/	0	3
Separation having length greater than 0.5 inch without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 5/	0	1
Premature function	0	1

1/ Misfiring sample cartridge(s), excepting those which occur in the waterproofness test (Table I), shall be retested as prescribed in 4.5.10. The lot shall not be penalized if the misfiring sample cartridge(s) fire in this test. If a misfiring cartridge fails to function in this test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.
b. Metal slivers across primer insulator.
c. Electrical continuity.
d. Missing or obstructed case vent.
e. Missing or insufficient primer composition or propellant.
f. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a is observed, however, the condition shall be reported for information. The "Acc" number shall be 0 and the "Rej" 3 for misfire(s) associated with b, c, or d. The second sample for evaluation of misfire(s) associated with condition b, c, or d shall consist of the function and casualty second sample (see Table I, Note 12/). The lot shall be rejected if either condition e or f is observed. Misfire(s) occurring in the waterproofness test shall be evaluated as specified in DARCOM-R 715-505, Volume 8.

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4.4.4 1 (Cont)

2/ Dwg. C7553663 (a detail of LI 7259063) shall apply for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

3/ Metal sheared or missing from the fired cartridge case exterior, such as a rim or neck shear, shall be classed as a defect. The lot shall not be penalized for metal shaved from the interior surface of the case neck in the crimped area.

4/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of a band fragment having no evidence of a band seat knurling imprint, and having length of 0.5 inch or less.

5/ Separation or breakup of projectile part(s), except rotating band separation, as witnessed by recovery of the part(s) of fragment(s), or hole(s) in the fragmentation screen(s), shall be classed as a defect.

4.4.4.2 Unlisted firing defects. The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test indicates that the product is unsuited for the purpose intended.

4.4.5 Packaging, packing, and marking inspection. Inspection for packaging, packing and container marking shall be in accordance with MIL-STD-644, as applicable.

4.4.6 Inspection equipment. Index of Inspection Equipment List LI 7259063 identifies the applicable inspection equipment required to perform examination and tests as specified herein. Equipment design(s) shall be in accordance with the applicable code designations. The code designations are defined on Dwg. B11075228, a detail of LI 7259063. The provisions of MIL-A-2550 shall apply.

4.5 Test methods and procedures.

4.5.1 Projectile torque. Equipment shall be as specified in LI 7259063. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque specified shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalignment in the scratch mark, shall be recorded.

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4.5.2 Projectile extraction. The method of test shall be as specified in DARCOM-R 715-505, Volume 8.

4.5.3 Propellant contamination. The propellant from each of the cartridges tested for projectile extraction shall be examined visually for contamination, as specified in DARCOM-R 715-505, Volume 8.

4.5.4 Electrical resistance.

4.5.4.1 Primed cartridge case. The equipment and circuitry specified in LI 7259063 shall be used to measure the resistance of the primer in each primed cartridge case.

4.5.4.2 Verification. A Simpson Volt-Ohm-Ammeter, M260, or approved equal, shall be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX100 and RX10,000 scales of the meter shall be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum.

4.5.5 Velocity, pressure and waterproofness. The methods of test shall be as specified in DARCOM-R 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed. (see 6.7)

4.5.6 Accuracy. The method of test and measurement of targets shall be as specified in DARCOM-R 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.5.7 Action time. The method of test shall be as specified in DARCOM-R 715-505, Volume 8.

4.5.8 Projectile function. The method of test shall be as specified in DARCOM-R 715-505, Volume 8. The aluminum target (2024T4, T3 or T36) of .040 inch nominal thickness shall be located 200 $\pm$ 25 yards from the gun muzzle normal to the line of fire. Observation of projectile functioning shall be made from the camera position.

4.5.9 Function and casualty. The method of test shall be as specified in DARCOM-R 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges. The weapon barrel shall be at ambient temperature at beginning of test and cooled to ambient after each 100 rounds. The cyclic rate of each burst shall not be less than 750 shots per minute (SPM) in the M24 gun. The cyclic rate for each burst shall be recorded. There shall be observation for a minimum distance of 400 feet from the gun muzzle for premature bursts.

4.5.10 Misfires. Misfiring cartridges shall be retested in the test equipment specified for the accuracy test. Results of examinations required by Table II shall be recorded.

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4.5.11 Defective weapon. If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. Any firing defect that occurred shall be valid for lot penalty in accordance with Table II unless the defect is found to have been caused by the defective weapon.

5. PACKAGING

5.1 Packing, level A. The type of ammunition pack shall be in accordance with instructions from the procuring activity. Linked ammunition shall be packed in accordance with Dwg. F7258967, F7259188, F7259189, or F11075755. Bulk ammunition shall be packed in accordance with Dwg. F7259119 or F7259122.

5.2 Marking. Marking on the shipping and storage container shall be in accordance with Dwg. D7258928 or D7259237, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements. (see 3.3, 4.3 and 6.3)

6.1.4 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.1.5 Provision for the supply, maintenance and disposition of Government furnished inspection equipment for acceptance inspection purposes.

6.1.6 Provision for submission and approval of the manufacturing process. (see 3.2)

6.1.7 Provisions for the inclusion of MIL-STD-1167, Ammunition Data Cards on DD form 1423, Contract Data Requirements List.

6.2 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical, chemical or physical properties of the item.

6.3 First article sample. In order to minimize inspection costs, the firing tests of 4.4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the testing facility.

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6.4 AQL's. The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation (see 4.4.2.3.c) is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.5 Combining tests. With approval of the procuring activity, tests may be performed concurrently on the same sample cartridges provided that the test results are not affected by this procedure. The test facility should make maximum use of this procedure in order to minimize testing costs.

6.6 Contamination of explosive components. Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfires, increase action time, and lot rejection.

6.7 Computations.

6.7.1 Standard deviation. Where computations of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$S = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}} \quad \text{or equivalent}$$

6.7.2 Range plan. Where a range plan is specified, the individual values will be recorded in subgroups of five in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range ($\bar{R}$).

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.9 Document identification change. AMCR 715-505, Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges, has been identified as DARCOM-R without change. (Pen and ink change AMCR to DARCOM-R.)

Custodian:

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Preparing Activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-46621C(AR) Crtg, 20mm Hi Xpl Incnd, M97A2

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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